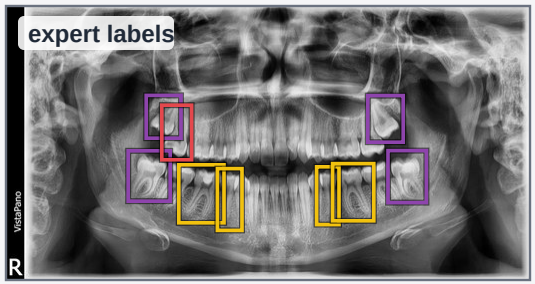
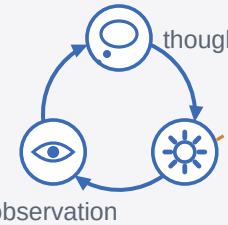
1 Generate demonstrations



impacted · deep caries · caries

Teacher loop with real tools



box perturbation

45	25	30

on the shown box

real tool outputs on one crop



zoom_ crop · window_ level · enhance_ contrast · denoise

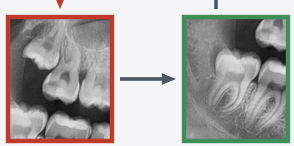
2 Verify and repair

Trace: DENTEX image 624
23 turns · 22 tool calls

T0	window_level — survey the full field	
T1	fdi_label — orient to FDI notation	
T2	locate_tooth(34) — box [1668, 917, 84, 322]	1
T3	zoom_crop — "only bone and root … too low"	2
T4	nudge_crop — dy –0.3 · scale 0.6	3
T6	nudge_crop — dy –0.5 · scale 0.7	3
T8	nudge_crop — dy –0.6 · scale 0.8	3
T9–10	zoom_crop + denoise — "… clearly see tooth 34"	
T14+	compare + enhance — contralateral, contrast	
T22	final answer — Q3 T4 · Caries · 0.82	4

1. **Format** — valid FDI notation and JSON
2. **No leakage** — no leaked generator directive
3. **Evidence** — claims backed by shown crops
4. **Diagnosis** — matches the expert label

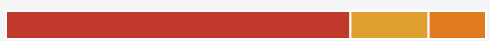
reject · re-verify · **97.5% pass first time**



drifted → repaired

editor repairs the trace, keeps the valid reasoning

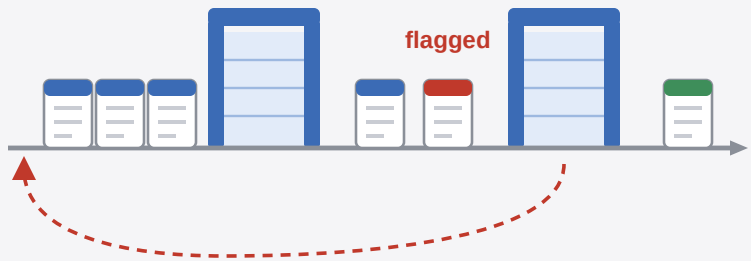
Why traces are rejected



- 66 invented findings
- 15 label contradictions
- 11 uncorrected drifts

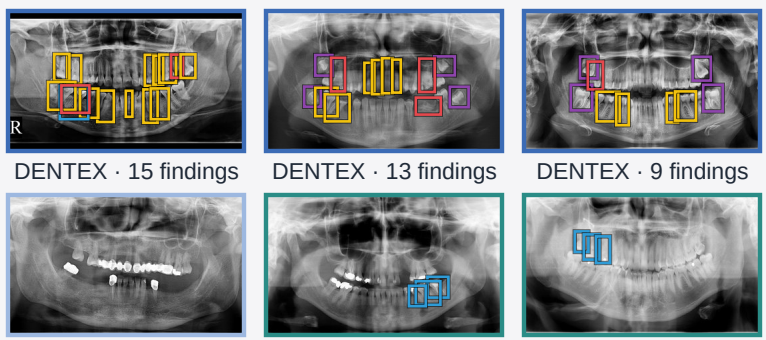
3 Decontaminate and release

regex sweep — leaked directives
re-verify flagged — traces (family B)



excise and regenerate

Released corpus: 3,694 traces



DENTEX · 15 findings · DENTEX · 13 findings · DENTEX · 9 findings
DENTEX · healthy scan · Tufts · 3 findings · Tufts · 3 findings

impacted · deep caries · caries · periapical

with tools		1,847
without tools	single turn	1,847